Concept Review
Radar

Why Use a Radar?

Radar provides reliable distance measurements using radio wave signals that can penetrate or reflect off common materials. It works in conditions where cameras or lidar may fail, such as darkness, partial occlusion or over long distance. It can be used to detect both large objects and tiny motions using phase signals.

Sensing Principle

This section will primarily focus on the sensing principle of pulsed radars, which slightly differs from Frequency-Modulated Continuous-Wave (FMCW) radars. "Pulsed" means the sensor transmits extremely short "pulse" of sinusoidal wave, called a wavelet. These pulses reflect off objects and return to the sensor after some time delay, t_{delay}. The Distance to an object can be calculated using:

$$d = \frac{c_0 t_{delay}}{2\sqrt{\varepsilon_r}}$$

where ε_r is the relative permittivity of the medium, c_0 is the speed of light, and the '2' in the denominator accounts for the round trip travel time of the radar pulse. The wavelets has a carrier frequency of 60.5 GHz which corresponds to a wavelength of roughly 5 mm. A 5 mm shift of the returned wavelet corresponds to only 2.5 mm of physical movement because of the round trip.

The radar integrated to Quanser's Mechatronics Sensor trainer is specifially a pulsed coherent radar. The term "coherent" means that the starting phase of the transmitted signal is known, as illustrated in Figure 1. As such the returned signal contains both amplitude and phase information, which can be useful in different applicatoins. For example, phase information can be utilized to detect very small movements or speed. Additionally, pulse timing is controlled by the Pulse Repetition Frequency (PRF). The sensor utilzied is a single-channel radar, which means it has 1 transmit antenna (TX), and 1 receive antenna (RX), allowing for the determination of distance but not angle to objects.

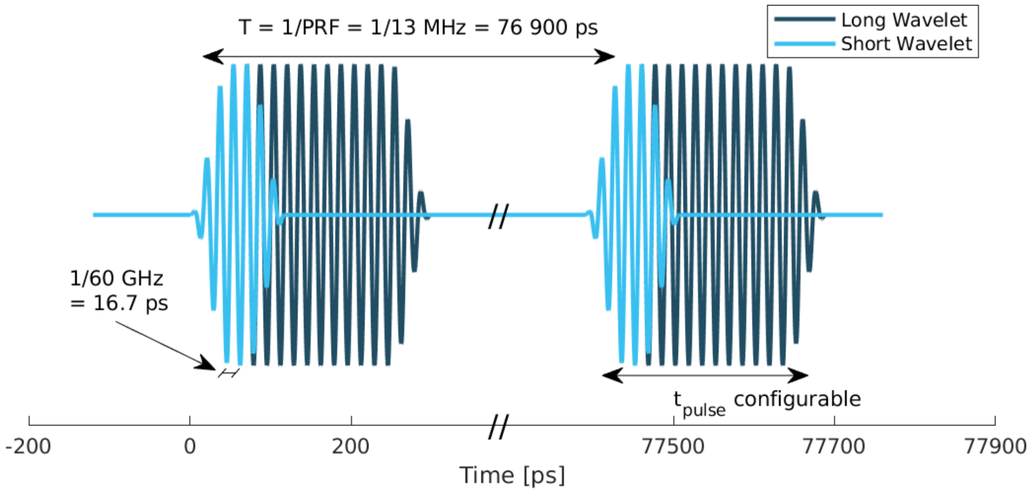

Figure 1. Illustration of the time domain transmitted signal from an Acconeer sensor using a PRF of 13 MHz. A radar sweep typically consists of thousands of pulses.

Reflectivity

When a radar wave encounters a boundary between two materials, part of the wave is reflected and part continues through the material. The amount of reflected energy depends on the difference in relative permittivity between the two media. The reflectivity of the object, γ, is given as:

$$\gamma = \left(\frac{\sqrt{\varepsilon_1} - \sqrt{\varepsilon_2}}{\sqrt{\varepsilon_1} + \sqrt{\varepsilon_2}}\right)^2$$

where ε_1 and ε_2 is the relative permittivity, at 60 GHz, on either side of the boundary. Keep in mind that the relative permittivity is generally frequency dependent and may also vary depending on the exact material composition and manufacturing process.

Materials with ε close to air, such as plastics and textile, reflect only a small portion of the incoming wave, which makes them partially transparent to 60 GHz radar. In contrast, materials such as water, skin, and especially metals have much larger permittivity differences, producing strong reflections. This is why objects like metal bottles or containers with water give large radar peaks, while empty plastic bottles or cardboard produce much weaker ones. As each boundary where permittivity changes produces a reflection, radar can even detect objects behind some materials. Table 1 lists approximate values for the real part of the relative permittivity for some common materials.

Table 1. Approximate relative permittivity of common materials

Material	Re(ε) at 60 GHz	with air boundary
Air	1	0
ABS	2.5-4.0	0.05 - 0.11
Polyethylene (PE)	2.3	0.042
Polypropylene (PP)	2.2	0.038
Polycarbonate	2.75	0.06
Mobile phone glass	6.9	0.2
Plaster	2.7	0.059
Concrete	4	0.11
Wood	2.4	0.046
Textile	2	0.029
Metal	–	1
Human skin	8	0.22
Water	11.1	0.28

In-phase and Quadrature (IQ)

While intuitively the amplitude of reflections can be used to estimate distances to objects, it does not have the capability to sense tiny movments. his is where phase information becomes powerful. Because the radar wavelength at 60 GHz is only about 5 mm, A 2π phase rotation of an IQ data point corresponds to a movement at the specific distance of $\lambda/2 \approx 2.5$ mm, providing high relative spatial resolution. Even a tiny movement of a few tenths of a millimeter causes a noticeable change in the phase of the returned signal. This means the radar can detect motions far too small to see in the amplitude alone, such as a person's breathing or subtle swaying when standing still.

The IQ service preserves this phase information by outputting the radar signal in complex form. Each reflection at a given distance can be represented as a phasor

$$x[d] = A_d e^{j\phi_d},$$

where $A_d$is the reflection strength and $\phi_d$is its phase. Using Euler's formula, the radar expresses this phasor as real and imaginary components:

$$x[d] = I[d] + jQ[d],$$

with $I[d] = A_d \cos(\phi_d)$and $Q[d] = A_d \sin(\phi_d)$. From these two values, students can compute the amplitude

$$A[d] = \sqrt{I[d]^2 + Q[d]^2}$$

and the phase

$$\phi[d] = \text{atan2}(Q[d], I[d]).$$

By providing stable I and Q data, this mode allows engineers to observe how the phase changes over time.

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